exchange data centers,[1449] for a total initial external cost of $414,250 per broker-dealer,[1450] and an aggregate estimate of 1,800 initial burden hours[1451] and $828,500 in initial external costs.[1452]

(ii) Comments/Responses on Initial Burden and Costs

In response to the commenter that believed that the estimated costs incurred by potential competing consolidators to build or upgrade systems to comply with proposed Rules 614(d)(1) through (d)(4) should be increased,[1453] the Commission is modifying its initial burden and cost estimates for broker-dealers that aggregate market data, as discussed below.

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is increasing its estimated initial costs and associated burden hours for broker-dealers that aggregate market data for internal use to modify their systems comply with Rules 614(d)(1) through (d)(4). The Commission preliminarily believed that the initial burden hour and external cost estimates for these broker-dealers to modify their systems to comply with

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1449 This estimate was based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

1450 $414,250 = [($206,250 in initial external costs to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 for the first month of market data costs) + ($194,000 in initial co-location costs at four exchange data centers)].

1451 The Commission estimates the monetized initial burden for this requirement to be $587,500. Based on discussions with a market participant, the Commission reached the following estimates: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] x [(2 broker-dealers)] = 1,800 initial burden hours across the broker-dealers.

1452 The Commission preliminarily estimates that broker-dealers would incur the following initial external costs: [($206,250 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers) x (2 broker-dealers)] = $828,500.

1453 IDS Letter I at 13, n.38.

Rules 614(d)(1) through (d)(4) would be similar to market data aggregation firms because, for both types of respondents, the scope of the systems changes and costs associated with becoming competing consolidators would be comparable. The Commission continues to believe this assumption is valid and is increasing its estimates for these broker-dealers as it is doing for market data aggregation firms. Most of these firms would have to spend substantial time coding for the new technical changes and would likely not have all of the components required to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission initially believed that competing consolidators would build aggregation systems in a single data center; however, the Commission now believes that competing consolidators may build systems for aggregating data in more than one data center. The Commission believes broker-dealers that aggregate market data would likely incur external costs greater than the Commission's estimate to buy new technology (for example, hardware and network infrastructure). The Commission is also revising its total initial burden hour and external cost estimates across all broker-dealers that aggregate market data to reflect a reduction in the number of potential competing consolidators that are broker-dealers that aggregate market data.

As it did for its market data aggregation firm estimates, the Commission is increasing its estimated burden hours for Sr. Programmers and Sr. Systems Analysts by three times as well as its estimated external costs to be incurred by broker-dealers that aggregate market data to purchase new technology to upgrade their systems to comply with Rules 614(d)(1) through (d)(4). The Commission estimates that each broker-dealer that aggregates market data would incur 2,200 initial burden hours to modify its systems to comply with Rules 614(d)(1) through

(d)(4),[1454] and initial external costs of $618,750 to purchase the necessary technology to effect such modifications,[1455] $194,000 to establish co-location connectivity to the exchange data centers,[1456] and $14,000 to purchase market data from the exchanges,[1457] for a total external cost to each broker-dealer that aggregates market data of $826,750.[1458] The Commission estimates that the total initial burden hours for one broker-dealers that aggregates market data would be 2,200 burden hours,[1459] and that total initial external costs would be $826,750 for one broker-

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1454 The Commission estimated the monetized initial burden for this requirement to be $697,150. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 1,050 hours) + (Sr. Systems Analyst at $285/hour for 900 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 2,200 initial burden hours to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1455 This estimate was originally based on discussions with a market participant and the Commission's understanding of hardware costs. The Commission has increased this estimated cost by three times because the Commission believes competing consolidators may potentially build aggregation systems in three data centers.

1456 This estimate is based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

1457 As it did in the Proposing Release, the Commission is using the monthly market data access and redistribution fees charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1458 The Commission estimated that a broker-dealer that aggregates market data would incur the following initial external costs: [($618,750 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] = $826,750.

1459 The Commission estimated the monetized initial burden for this requirement to be $697,150. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 1,050 hours) + (Sr.

dealer that aggregates market data to modify its systems to comply with Rules 614(d)(1) through (d)(4).[1460]

(d) Initial Burden and Costs for Exclusive SIPs

(i) Proposed Estimates – Initial Burden and Costs

In the Proposing Release, the Commission estimated that the exclusive SIPs may have to make a greater scope of changes to become competing consolidators than market data aggregation firms. For this reason, the Commission estimated initial burden hour and external cost estimates that were higher than those estimated for market data aggregation firms.[1461] Specifically, the Commission estimated that each exclusive SIP would incur 1,800 initial burden

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Systems Analyst at $285/hour for 900 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] x [(1 broker-dealer that aggregates market data)] = 2,200 total initial burden hours.

1460 The Commission estimated that broker-dealers that aggregate market data would incur the following total initial external costs: [($618,750) to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] x [(1 broker-dealer that aggregates market data)] = $826,750.

1461 In the Proposing Release, the Commission doubled its initial burden hour and external cost estimates for a market data aggregation firm to reach its initial burden hour and external cost estimates for an exclusive SIP.

hours[1462] and $412,500 in external costs[1463] to modify its systems to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission estimated that an exclusive SIP would incur initial external costs of $14,000 to purchase market data from the SROs,[1464] and an additional initial external cost of $194,000 to co-locate itself at four exchange data centers,[1465]

1462 Based on discussions with a market participant, the Commission reached the following estimates for a market data aggregation firm: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 6 months (900 burden hours) to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for a 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead. As noted above, the Commission increased this initial burden hour estimate for the exclusive SIPs. Therefore, the Commission estimated that each exclusive SIP will incur 1,800 initial burden hours to upgrade its existing systems to comply with Rules 614(d)(1) through (d)(4) (or $587,500, as monetized).

1463 As noted above, the Commission estimated the initial external cost estimates to comply with Rules 614(d)(1) through (d)(4) to be higher for exclusive SIPs than for market data aggregation firms. The Commission estimated that each exclusive SIP will incur $412,500 in initial external costs to modify its systems to comply with Rules 614(d)(1) through (d)(4).

1464 The Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1465 This estimate was based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

for a total initial external cost of $620,500 per existing exclusive SIP,[1466] and an aggregate estimate of 3,600 initial burden hours[1467] and $1,241,000 in initial external costs.[1468]

(ii) Comments/Responses on Initial Burden and Costs

In response to the commenter that believed that the estimated costs incurred by potential competing consolidators to build or upgrade systems to comply with proposed Rules 614(d)(1) through (d)(4) should be increased,[1469] the Commission is modifying its initial burden and cost estimates for the exclusive SIPs, as discussed below.

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is increasing its estimated initial costs and associated burden hours for exclusive SIPs that choose to become competing consolidators to upgrade their systems to

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1466 The Commission estimated that each exclusive SIP would incur the following initial external costs: [($412,500 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers)] = $620,500.

1467 Based on discussions with a market participant, the Commission reached the following estimates for a market data aggregation firm: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 900 initial burden hours across the market data aggregation firms. As noted above, the Commission increased this initial burden hour estimate to apply to the exclusive SIPs. Therefore, the Commission preliminarily estimated that each exclusive SIP will incur 1,800 initial burden hours to upgrade its existing systems to comply with Rules 614(d)(1) through (d)(4) (or $587,500, as monetized). The aggregate initial burden hour estimate for two exclusive SIPs would be [(1,800 initial burden hours) x (2 exclusive SIPs)] = 3,600 initial burden hours.

1468 The Commission preliminarily estimated that the exclusive SIPs would incur the following initial external costs: [($412,500 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers)] x [(2 exclusive SIPs)] = $1,241,000.

1469 IDS Letter I at 13, n.38.

comply with Rules 614(d)(1) through (d)(4). The Commission preliminarily believed that the exclusive SIPs would have to make a greater scope of changes to become competing consolidators than market data aggregation firms. For this reason, the Commission preliminarily estimated initial burden hour and external cost estimates that were higher than those estimated for market data aggregation firms.[1470] The Commission continues to believe that exclusive SIPs will have to make greater changes to their systems than market data aggregation firms to comply with Rules 614(d)(1) through (d)(4). However, like market data aggregation firms, exclusive SIPs will have to spend substantial time coding for the new technical changes and would likely not have all of the components required to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission initially believed that competing consolidators would build aggregation systems in a single data center; however, the Commission now believes that competing consolidators may build systems for aggregating data in more than one data center. The Commission believes exclusive SIPs would likely incur external costs greater than the Commission's estimate to buy new technology (for example, hardware and network infrastructure).

As it did for its market data aggregation firm estimates, the Commission is increasing its estimated burden hours for Sr. Programmers and Sr. Systems Analysts employed by each exclusive SIP by three times, as well as its estimated external costs to be incurred by the exclusive SIPs to purchase new technology to upgrade their systems to comply with Rules 614(d)(1) through (d)(4). The Commission estimates that each exclusive SIP would incur 4,400

[1470] See supra note 1461.

initial burden hours to modify its systems to comply with Rules 614(d)(1) through (d)(4),[1471] and initial external costs of $1,237,500 to purchase the necessary technology to effect such modifications,[1472] $194,000 to establish co-location connectivity to the exchange data centers,[1473] and $14,000 to purchase market data from the exchanges,[1474] for a total external cost to each exclusive SIP of $1,445,500.[1475] The Commission estimates that the total initial burden hours for two exclusive SIPs would be 8,800 burden hours,[1476] and that total initial external costs would be

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1471 The Commission estimated the monetized initial burden for this requirement to be $1,394,300. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 2,100 hours) + (Sr. Systems Analyst at $285/hour for 1,800 hours) + (Compliance Manager at $310/hour for 200 hours) + (Director of Compliance at $489/hour for 100 hours) + (Compliance Attorney at $366/hour for 200 hours)] = 4,400 initial burden hours to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1472 This estimate was originally based on discussions with a market participant and the Commission's understanding of hardware costs. The Commission has increased this estimated cost by three times because the Commission believes competing consolidators may potentially build aggregation systems in three data centers.

1473 This estimate is based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

1474 As it did in the Proposing Release, the Commission is using the monthly market data access and redistribution fees charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1475 The Commission estimated that each exclusive SIP would incur the following initial external costs: [($1,237,500 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] = $1,445,500.

1476 The Commission estimated the monetized initial burden for this requirement to be $1,394,300. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 2,100 hours)

$2,891,000 for two exclusive SIPs to modify their systems to comply with Rules 614(d)(1) through (d)(4).[1477]

(e) Initial Burden and Costs for New Entrants

(i) Proposed Estimates – Initial Burden and Costs

In the Proposing Release, the Commission estimated that each new entrant would incur 3,600 initial burden hours[1478] and $825,000 in external costs[1479] to build systems to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission estimated that a new entrant

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+ (Sr. Systems Analyst at $285/hour for 1,800 hours) + (Compliance Manager at $310/hour for 200 hours) + (Director of Compliance at $489/hour for 100 hours) + (Compliance Attorney at $366/hour for 200 hours)] x [(2 exclusive SIPs)] = 8,800 initial burden hours across the exclusive SIPs.

1477 The Commission estimated that the exclusive SIPs would incur the following initial external costs: [($1,237,500 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] x [(2 exclusive SIPs)] = $2,891,000.

1478 Based on discussions with a market participant, the Commission reached the following estimates for a market data aggregation firm: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 6 months (900 burden hours) to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead. As noted above, the Commission increased this initial burden hour estimate to apply to the new entrants. Therefore, the Commission estimated that each new entrant would incur 3,600 initial burden hours to build systems to comply with Rules 614(d)(1) through (d)(4) (or $1,175,000, as monetized).

1479 As noted above, the Commission increased its initial external cost estimates for market data aggregation firms to apply to new entrants. In particular, the Commission estimated that each new entrant will incur $825,000 in initial external costs to build systems to comply with Rules 614(d)(1) through (d)(4).

would incur initial external costs of $14,000 to purchase market data from the SROs,[1480] and an additional initial external cost of $194,000 to co-locate itself at four exchange data centers,[1481] for a total initial external cost of $1,033,000 per new entrant,[1482] and an aggregate estimate of 7,200 initial burden hours[1483] and $2,066,000 in initial external costs.[1484]

(ii) Comments/Responses on Initial Burden and Costs

In response to the commenter that believed that the estimated costs incurred by potential competing consolidators to build or upgrade systems to comply with proposed Rules 614(d)(1)

1480 The Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1481 This estimate was based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

1482 The Commission estimated that each new entrant would incur the following initial external costs: [($825,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers)] = $1,033,000.

1483 Based on discussions with a market participant, the Commission reached the following estimates for a market data aggregation firm: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 900 initial burden hours. As noted above, the Commission increased the per market data aggregation firm initial burden hour estimate to apply to the new entrants. The Commission estimated that each new entrant would incur 3,600 initial burden hours to build systems to comply with Rules 614(d)(1) through (d)(4) (or $1,175,000, as monetized). [(3,600 burden hours) x (2 new entrants] = 7,200 hours (or $2,350,000 as monetized).

1484 The Commission estimated that each new entrant would incur the following initial external costs: [($825,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers) x (2 new entrants)] = $1,033,000. [($1,033,000 in initial external costs) x (2 new entrants)] = $2,066,000.